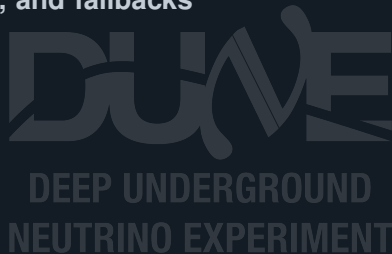

DAPHNE Production Readiness Review

A reviewer-facing map of evidence, risks, decisions, and fallbacks



Manuel Arroyave

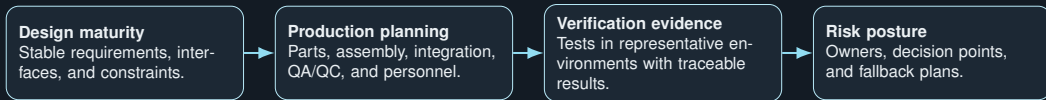
Draft, May 2026

SECTION

Review Decisions First

Judge what is ready for production, what remains a controlled risk, what is out of scope, and what evidence closes each risk.

What The Review Panel Should Decide



Decision aid

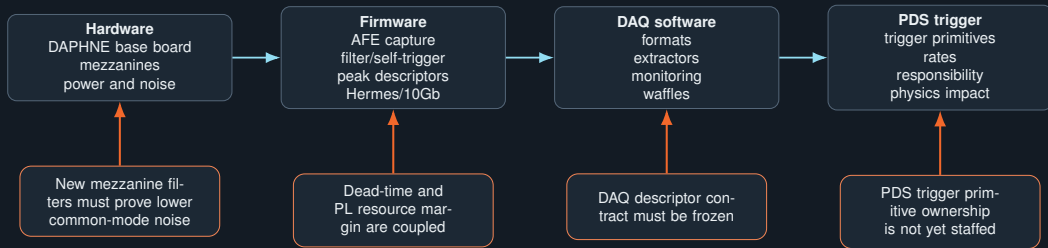
The PRR package should give the panel a readiness matrix, not only schematics, branches, plots, and status notes.

What to look for

The material should answer production questions directly: what passes, what is open, who owns it, and what is deliberately out of scope.

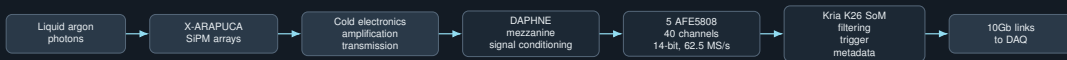
Source: Public PRR definitions emphasize design readiness, production planning, integration support, personnel, and unacceptable cost/schedule/performance risk.

Review The Handoffs, Not Only The Board



Review focus: DAPHNE becomes production-ready only when the hardware, firmware, DAQ, and trigger contracts close together.

DAPHNE In Plain Language For Reviewers



Analog problem

Preserve signal fidelity and reduce noise before digitization.

Firmware problem

Keep trigger sensitivity without exceeding K26 PL resources.

DAQ problem

Make descriptors usable by monitoring, analysis, and trigger primitive builders.

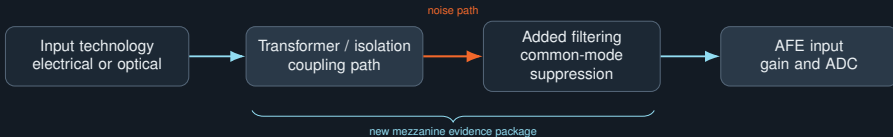
Source: DAPHNE public electronics descriptions identify plug-in mezzanines, AFE5808 digitization, K26 processing, trigger/metadata calculation, and high-speed DAQ links.

SECTION

Keep Scope Sharp

Separate production hardware, integration work, and architecture studies before the review drifts into every open R&D thread.

Hardware: Evidence To Close Noise Risk



Evidence to request

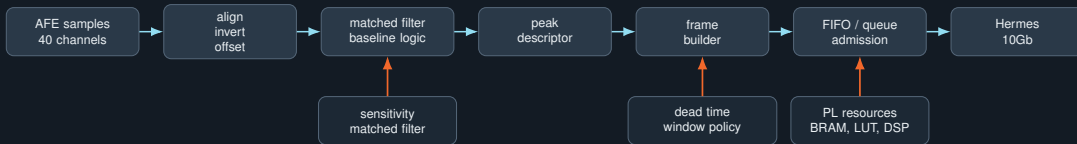
Schematic/BOM delta and before/after noise spectra with the same fixture.

Acceptance criterion

Pass/fail tied to PDS S/N and threshold, plus a production QA test for assembly drift.

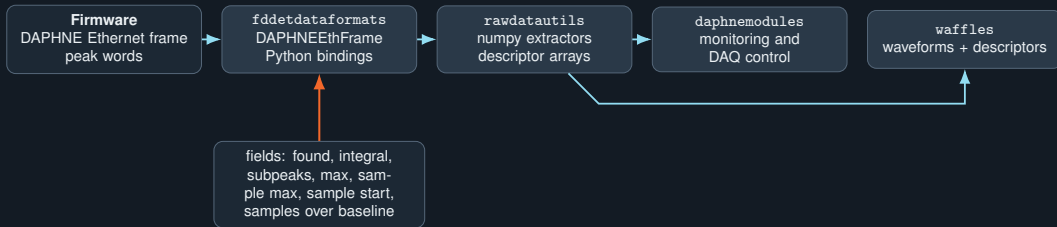
Status: mitigation under qualification until the noise data is in the evidence matrix.

Firmware: What Tradeoff Is Under Review?



Reviewer translation: this is not one algorithmic choice. Sensitivity, dead time, DAQ compatibility, and K26 fit move together.

DAQ: What Contract Is Ready To Freeze?



ready

Format support exists on DAPHNE ETH branches.

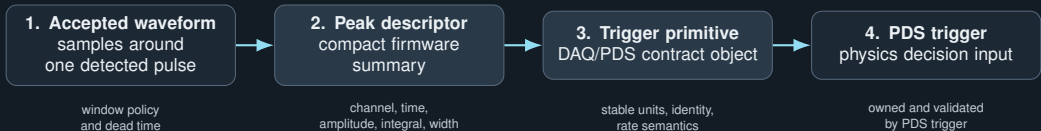
in work

Monitoring and DAQ run integration still need clean review evidence.

at risk

Trigger primitive ownership is not solved by data extraction alone.

Peak Descriptors: What Must Be Owned?



What to accept

The descriptor is the firmware product. The trigger primitive is the DAQ/PDS contract built from that product.

What to challenge

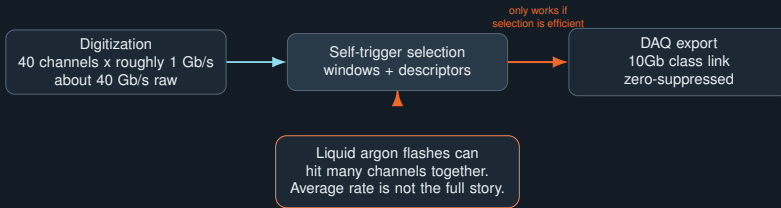
Dead-time rejects create missing primitives; descriptor semantic drift creates wrong primitives; both need an owner.

SECTION

Judge Dead Time Carefully

The problem is not just bandwidth or code. It is a rate, buffering, and contract problem.

Why Dead Time Is A Review Risk



What not to assume

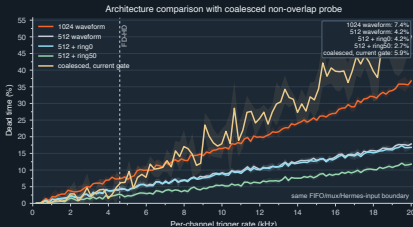
Dead time is set by admission, buffering, merging, and release, not by Ethernet bandwidth alone.

What to require

Do not accept “below 2%” until the flash-correlation model and operating point are accepted by DAQ/PDS.

Simulation: Dead-Time Options

Model	4.6 kHz/ch	10 kHz/ch	14 kHz/ch	20 kHz/ch
1024 waveform	7.41%	16.46%	24.02%	36.76%
512 waveform	4.20%	9.14%	12.52%	17.83%
512 + ring50	2.65%	5.47%	7.55%	11.77%
coalesced, current gate	5.92%	18.06%	27.37%	50.22%
coalesced, relaxed gate	0.01%	0.33%	0.46%	1.03%



Reviewer reading

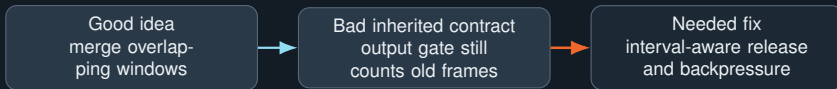
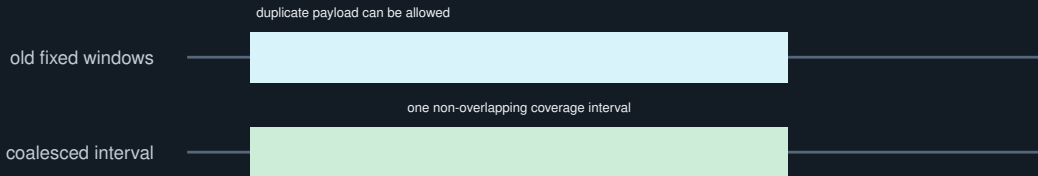
512 windows are the first robust improvement.

Ring50 is the best fixed-record curve.

Coalesced only becomes excellent after the output gate changes.

More lanes are secondary to the window admission contract.

Coalesced Transport: Not Yet A Production Claim



Coalescing removes duplicate waveform transport, but it only reduces dead time after the downstream frame/queue contract stops behaving like every accepted trigger must reserve a full independent fixed record.

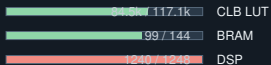
K26 Resources: Does The Design Have Margin?

Known build references

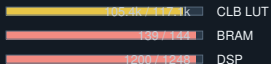
- ▶ a389fcd: routed clean baseline.
- ▶ 27a4ca9: ring2k routed clean, but BRAM tight.
- ▶ grouped Hermes draft: placement failed on BRAM.
- ▶ xcorr DSP reduction: candidate to trade DSP for CLB/carry logic.

Resource bars

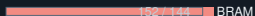
Baseline a389fcd



Ring2k 27a4ca9



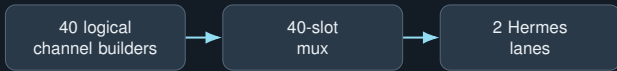
Grouped draft



Dead-time mitigation competes directly with BRAM/LUT/DSP margin. The next architecture must remove replication deliberately, not add another layer of FIFOs.

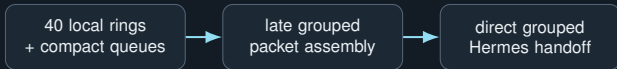
Transport: Production Versus Study

Current production-compatible shape



Good for compatibility
but expensive replication

Architecture study target



Better architecture
but must prove
BRAM and dead time

Design principle: do not jump to 5 AFE producers just because it is cheaper. It risks cross-channel blocking. Prefer local channel storage with late grouped assembly sized by simulation and BRAM budget.

SECTION

Reviewable Claims

Every production claim should map to evidence, an owner, and a fallback.
That is the PRR package.

Which Claims Are Ready For Review?

Production reference: ready

The routed firmware baseline and overlay packaging path exist. Keep this as the comparison point for any mitigation branch.

Mezzanine noise filters: evidence gap

The mitigation is credible, but production readiness needs before/after spectra, a schematic/BOM delta, and an acceptance test.

DAQ descriptor chain: in work

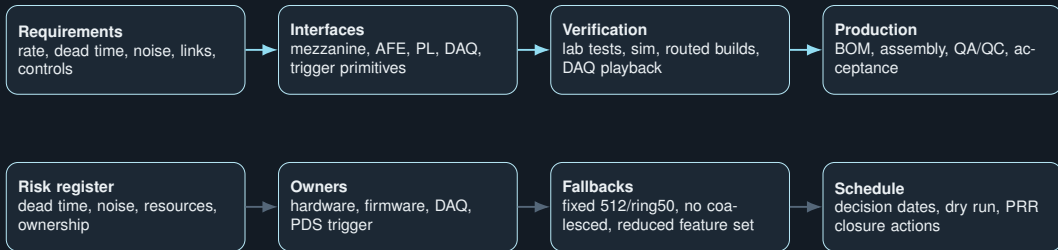
Format bindings, raw-data extraction, monitoring, and waffles integration exist on active branches. The missing PRR item is a reviewed release path with tests.

Dead time and PDS trigger: at risk

Simulations clarify the choices, but the production architecture and trigger primitive owner group are still open risks.

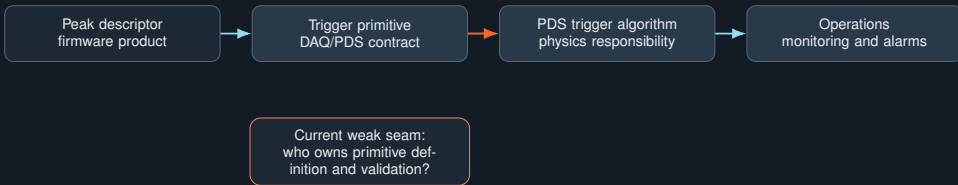
Backups keep the detailed source map; the main deck stays decision-oriented.

Evidence Packet The Panel Should Expect



The PRR deck should point to one controlled evidence matrix, not ask reviewers to infer status from many branches.

PDS Trigger Ownership: The Risk To Challenge



What we have

Firmware can produce peak descriptors, and DAQ can expose them to monitoring and analysis.

What is still missing

A named owner group and acceptance tests for descriptor-to-trigger-primitive behavior.

Is This Ready To Become The PRR Package?

Yes, as a draft

The existing presentations, firmware docs, dead-time simulations, DAQ branches, and hardware schematics are enough to draft the PRR story and identify the main open risks.

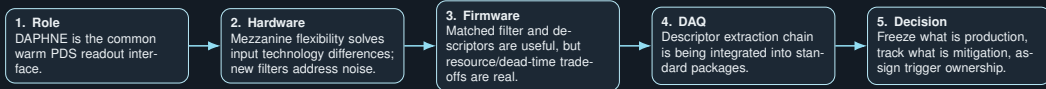
No, not as the final review package

It is not clean enough yet for DOE reviewers because the evidence is spread across repos and branches, and some fronts are status notes rather than acceptance criteria.

Reviewer-facing cleanup

- ▶ One readiness matrix with owner, evidence, status, and next action.
- ▶ One interface-control view from AFE to DAQ trigger primitives.
- ▶ One dead-time requirement and one accepted operating model.
- ▶ One hardware noise validation plan for the mezzanine filters.
- ▶ One DAQ release path for descriptor extraction and monitoring.

Narrative The Panel Can Follow



Main message for reviewers: DAPHNE is close enough for a serious PRR package, if dead time and trigger primitives are presented as controlled production risks with owners and fallbacks.

Closure Plan Before Review



Surface risk early

Dead-time and ownership are better shown as controlled risks than discovered by reviewers.

Credible fallback

Fixed 512/ring50 is the compatibility fallback while coalesced transport matures.

Make it testable

Every claim should point to one plot, report, run, or package test.



SECTION

Backup

Sources, resource glossary, and detailed references for technical questions.



Backup: Source Map

Front	Local source	What it contributes
Firmware build and resources	<code>daphne-firmware/docs</code>	routed baseline, packaging boundary, architecture drafts, K26 constraints
Dead-time simulations	<code>daphne_mezz_xc_sim/docs</code>	C++ and RTL dead-time studies, ring/coalesced comparisons
Legacy mezzanine firmware	<code>Daphne_MEZZ</code>	original DAPHNE V3/MEZZ PL and Vivado script reference
Data format	<code>fddetdataformats</code>	DAPHNE Ethernet frame and peak descriptor bindings
Data extraction	<code>rawdatautils</code>	numpy arrays for waveform and peak descriptor fields
Monitoring / control	<code>daphnmodules</code>	DAQ control and online monitoring integration
Analysis	<code>waffles</code>	waveform/descriptor attachment for analysis workflows

Backup: Resource Glossary

CLB / LUT

General FPGA logic. Used by state machines, muxes, counters, control logic, adders, and shift/add filter implementations.

BRAM / URAM

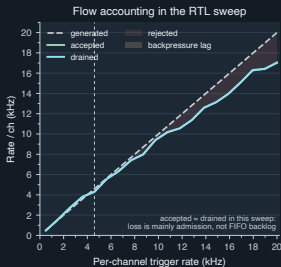
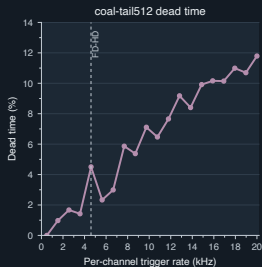
On-chip memories. Used by waveform rings, FIFOs, spy buffers, queues, and large buffers.

DSP48E

Dedicated arithmetic slices. Used by multiply/filter-heavy paths; the matched filter used to consume most of them.

If one resource class is above roughly 90–95%, small design changes can break placement, timing, or routing even when the HDL is functionally correct.

Backup: Dead-Time Takeaway



Plain-English summary

- ▶ 1024-sample records waste too much acceptance.
- ▶ 512-sample records are a safe first cut.
- ▶ Ring50 improves dead time while keeping the fixed-record contract.
- ▶ Coalesced transport is the long-term clean idea, but only if the output queue semantics change too.